DAY – 3 IPv4, IPv6 and Subnetting Assignment

1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

Ans:-

IPv4	IPv6
32-bit address	128-bit address
Written in dotted decimal, e.g. 192.168.1.1	Written in hexadecimal with colons, e.g. 2001:db8::1
About 4.3 billion addresses, which are running out	About 3.4 x 10^38 addresses, practically unlimited
Header is 20 bytes or more (variable size)	Header is a fixed 40 bytes, simpler for routers
Address set manually or by DHCP; NAT is widely used to save addresses	Address can be set automatically (SLAAC); NAT is not needed
Uses broadcast to reach all devices	No broadcast; uses multicast and anycast instead

2. Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Ans:-

Device	IP address	Subnet mask	Default gateway
Laptop	192.168.10.11	255.255.255.0	192.168.10.1
Phone	192.168.10.12	255.255.255.0	192.168.10.1
Smart TV	192.168.10.13	255.255.255.0	192.168.10.1
Printer	192.168.10.14	255.255.255.0	192.168.10.1

* The router (default gateway) uses 192.168.10.1. Every device has a different address in the same network 192.168.10.0/24, and none of them uses .0 (network address) or .255 (broadcast address). The static addresses should be kept outside the router's DHCP range so that they do not clash with other devices.

3. Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

Ans: -

- **Step 1:** /27 means 27 network bits, so the subnet mask is 255.255.255.224 (last octet 11100000 = 128 + 64 + 32 = 224).
- **Step 2:** The network 192.168.1.0 is a /24, so the bits borrowed for subnets = 27 - 24 = 3. Number of subnets = 2^3 = 8.
- **Step 3:** Host bits = 32 - 27 = 5. Hosts per subnet = 2^5 - 2 = 30 usable hosts.
- **Step 4:** Block size = 256 - 224 = 32, so the subnets start at 0, 32, 64, 96 and so on.

No.	Network address	First usable host	Last usable host	Broadcast address
1	192.168.1.0	192.168.1.1	192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33	192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65	192.168.1.94	192.168.1.95

* **Valid subnets: 8 (192.168.1.0, .32, .64, .96, .128, .160, .192, .224)**

Hosts per subnet: 30

Note: Older rules did not allow the first and last subnets, which would give 6 subnets. Modern networks use all 8.

4. You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

Ans: -

- **Step 1 (CIDR):** 255.255.255.0 has 24 bits set to 1 (8 + 8 + 8), so the CIDR notation is /24.
- **Step 2 (Network address):** Network address = IP AND mask. The first three octets stay the same and the last octet becomes 0, so 10.0.5.17 becomes 10.0.5.0.
- **Step 3 (Broadcast address):** Set all 8 host bits to 1. The last octet becomes 11111111 = 255, so the broadcast address is 10.0.5.255.
- **Step 4 (Valid hosts):** First host = 10.0.5.1 and last host = 10.0.5.254. Number of hosts = 2^8 - 2 = 254.

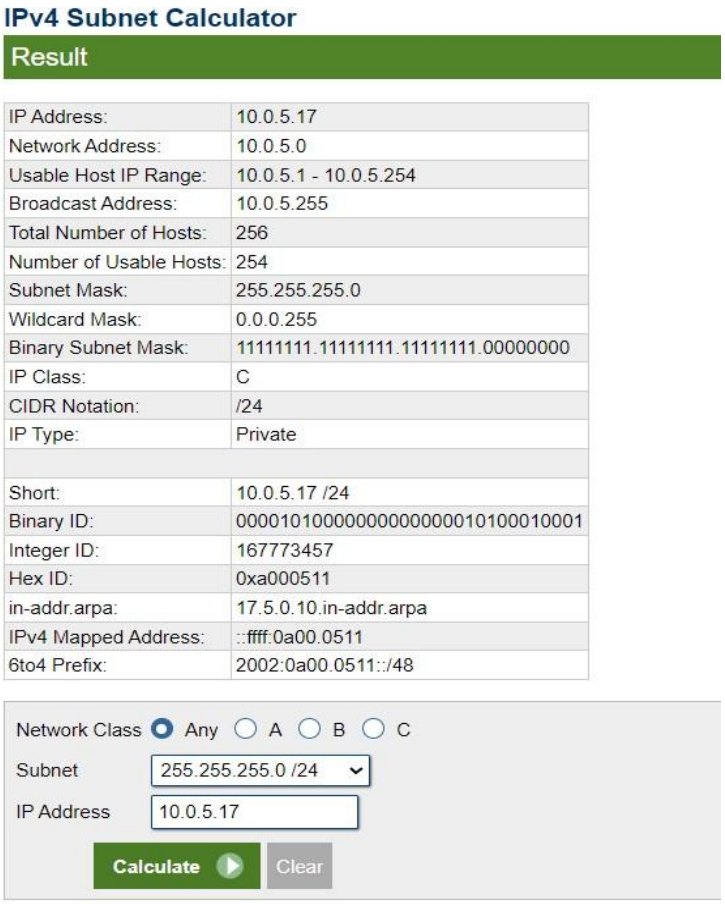
Step	Octet 1	Octet 2	Octet 3	Octet 4	Result
IP address (binary)	00001010	00000000	00000101	00010001	10.0.5.17
Subnet mask (binary)	11111111	11111111	11111111	00000000	255.255.255.0
Network = IP AND mask	00001010	00000000	00000101	00000000	10.0.5.0
Broadcast = host bits all 1	00001010	00000000	00000101	11111111	10.0.5.255

- **Network address: 10.0.5.0 (10.0.5.0/24)**
- **Broadcast address: 10.0.5.255**
- **Valid host range: 10.0.5.1 to 10.0.5.254 (254 hosts)**

5. Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

Ans: -

- **Steps:** Open the subnet calculator website, enter the IP address 10.0.5.17, and choose the subnet mask 255.255.255.0 (/24), and click Calculate. Then note the network address, broadcast address and usable host range, and take a screenshot.



IPv4 Subnet Calculator

Result

IP Address:	10.0.5.17
Network Address:	10.0.5.0
Usable Host IP Range:	10.0.5.1 - 10.0.5.254
Broadcast Address:	10.0.5.255
Total Number of Hosts:	256
Number of Usable Hosts:	254
Subnet Mask:	255.255.255.0
Wildcard Mask:	0.0.0.255
Binary Subnet Mask:	11111111.11111111.11111111.00000000
IP Class:	C
CIDR Notation:	/24
IP Type:	Private
Short:	10.0.5.17 /24
Binary ID:	00001010000000000000010100010001
Integer ID:	167773457
Hex ID:	0xa000511
in-addr.arpa:	17.5.0.10.in-addr.arpa
IPv4 Mapped Address:	::ffff:0a00.0511
6to4 Prefix:	2002:0a00.0511::/48

Network Class (•) Any ( ) A ( ) B ( ) C

Subnet 255.255.255.0 /24

IP Address 10.0.5.17

Calculate Clear

Item	My manual answer	Tool output
Network address	10.0.5.0	10.0.5.0 (matches)
Broadcast address	10.0.5.255	10.0.5.255 (matches)
Usable host range	10.0.5.1 - 10.0.5.254	10.0.5.1 - 10.0.5.254 (matches)
Usable hosts	254	254 (matches)

* **Conclusion:** My manual answer matched the tool's output: network 10.0.5.0, broadcast 10.0.5.255 and usable hosts 10.0.5.1 to 10.0.5.254.